

**Project I Project Report: Probabilistic Evaluation of Vendor Test
Accuracy**

EECS 331 – Probability and Statistics for Engineering Majors

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Introduction

This project applies probability theory to evaluate the reliability of a vendor's device testing process, where Event A denotes that a device is not defective and Event B denotes that the test reports the device as defective. The vendor provides four conditional probabilities relating these events, and our task is to determine whether these values are internally consistent with the axioms of probability, the law of total probability, and the complement rules. Because management is weighing a \$20 million investment on the basis of this data, even a small inconsistency in the conditional or joint probabilities could undermine the credibility of the entire dataset. By translating the vendor's claims into formal joint, marginal, and union probabilities and then testing them against the probability framework, we can objectively decide whether the data can be trusted or whether a correction is required before moving forward.

Background

- Sample Space (Ω): set of all possible outcomes of an experiment.
- Event: subset of Ω ; here, A = "device not defective," B = "test says defective."
- Complement Rule: $P(A^c) = 1 - P(A)$.
- Partition Identity: $\Omega = (A \cap B) \cup (A^c \cap B) \cup (A \cap B^c) \cup (A^c \cap B^c)$.
- Additivity: disjoint events' probabilities add.
- Conditional Probability: $P(E|F) = P(E \cap F) / P(F)$, $P(F) > 0$.
- Law of Total Probability: probabilities must sum across a partition to 1.
- Consistency Check: For any event B,
 $P(A|B) + P(A^c|B) = 1$.

- Complement Rule:

$$P(A^c) = 1 - P(A)$$

- Additivity (disjoint events):

$$P(E_1 \text{ or } E_2 \text{ or } \dots \text{ or } E_n) = \text{Sum } P(E_i)$$

- Partition Identity:

$$\Omega = (A \text{ and } B) \text{ or } (A^c \text{ and } B) \text{ or } (A \text{ and } B^c) \text{ or } (A^c \text{ and } B^c)$$

- Conditional Probability:

$$P(E|F) = P(E \text{ and } F) / P(F), \quad P(F) > 0$$

- Law of Total Probability:

$$P(A) = P(A | B) * P(B) + P(A | B^c) * P(B^c)$$

- Consistency Check (for any event B):

$$P(A|B) + P(A^c|B) = 1$$

Homework: Project I

Twenty years of their data from a firm wanting to sell us devices, shows:

- ① 90% of their devices are not defective when the tests say they are not.
- ② 5% of their devices are not defective when the tests say they are.
- ③ 85% of their devices are defective when the tests say they are.
- ④ 10% of their devices are defective when the tests say they are not.

Let A be the event that the device is not defective and let B be the event the tests says the device is defective. Compute the following:

- ① $P[A \cap B]$, $P[A \cap B^c]$, $P[A^c \cap B]$ and $P[A^c \cap B^c]$
- ② $P[A \cup B]$, $P[A \cup B^c]$, $P[A^c \cup B]$ and $P[A^c \cup B^c]$.
- ③ $P[A]$, $P[B]$, $P[A^c]$ and $P[B^c]$.
- ④ $P[B|A]$, $P[B|A^c]$, $P[A|B]$ and $P[A^c|B]$.

Problem

The problem is to determine, using the definitions and axioms of probability, whether these conditionals are internally consistent, whether the implied joint and marginal probabilities make sense, and ultimately whether this data can be trusted to support a twenty-million-dollar business decision.

Methodology

The exact sequence I used when moving between definitions to applications is as follows: First, I made sure the events were clearly defined in terms of the sample space and event space. Then, I noticed I could express each of the four vendor statements as conditional probability, using the definition that the probability of E given F equals the probability of E and F divided by

the probability of F. From there, I converted each conditional into a joint probability by multiplying the conditional value by the probability of the conditioning event, as outlined in my note examples on conditional probability. Once the four joint probabilities were written, the partition identity was applied, which is highlighted as a direct consequence of the additivity axiom, since the four joint events are disjoint and together exhaust the sample space. This step exposed the normalization issue immediately. To strengthen my evaluation, I also considered the complement rule, since it was stressed in class that the probability of an event and its complement must always sum to one. Finally, the consistency check from the notes was applied to the conditional probabilities, which made the contradiction in the vendor's claims undeniable. With my calculated approach, I demonstrated rigorously that the vendor's dataset is mathematically inconsistent and cannot be trusted for decision-making.

Analysis

Any random experiment is modeled with a sample space, an event space, and a probability function that must obey three axioms: non-negativity, normalization, and additivity. In this problem, the sample space is defined by device condition and test outcome, and the events are A, the device is not defective; A complement, the device is defective; B, the test reports defective, and B complement, the test reports not defective. It is emphasized that events of this form create a partition of the sample space, meaning their probabilities must sum to one, and any inconsistencies can be caught using the additivity axiom. The vendor provides four conditional probabilities, which are intended to describe the interaction between the test and device outcomes, but when these values are checked against the axioms, they fail. From the law of total probability, the four joint events A and B, A complement and B, A and B complement, and A

complement and B complement should together add to one. Yet when the vendor's numbers are substituted, the only way for the sum to equal one is if the test never reports defective at all, which contradicts two of the vendor's own statements. Even more directly, my notes call for the consistency check: for any conditioning event, the conditional probability of an event and its complement must add to one. The vendor reports that the probability of A given B is five percent and the probability of A complement given B is eighty-five percent, which sum to ninety percent rather than one hundred. This violates the conditional probability definition itself, showing that the data cannot be internally consistent with the rules of probability.

Results

The results of the evaluation show that the vendor's statements do not form a valid probability model. When the conditional probabilities were expressed as joint probabilities, the partition identity required them to add up to one. Instead, applying the additivity axiom forced the probability of the test reporting defective to be zero, which is not consistent with the given statements. The law of total probability also breaks down because the joint events cannot normalize properly. The consistency check makes the issue more direct: the probability of not defective given defective and the probability of defective given defective only add to ninety percent, while the definition of conditional probability requires that these values add to one hundred percent. This means the sample space and event space implied by the vendor's data are

not coherent. The complement rule and other axioms cannot be satisfied, so the provided values are internally inconsistent and cannot represent a true probability space.

Discussion

What stands out about the vendor's data is not only that it fails the rules of probability but also how easily the issue hides beneath numbers that seem reasonable at first glance. Each percentage on its own looks believable, yet when placed into the sample space and event space the structure does not hold together. This shows the importance of treating conditional probability with care, because even a small reporting mistake can disrupt the entire model. The gap revealed by the consistency check may come from a miscalculation or from confusing a conditional probability with a marginal probability. It is also possible that one of the values was recorded incorrectly, which would explain why the partition identity and the complement rule do not balance as expected. What this suggests is that the numbers themselves are not the problem, but rather the way they interact within the probability space. For decision-making, this creates a serious challenge because a dataset that cannot satisfy the law of total probability cannot be relied upon to give an accurate view of the experiment. The key lesson is that probability is not only about calculating numbers, it is about ensuring that the entire structure of events and their relationships is sound. When that structure breaks down, no individual figure can make the model trustworthy.

Conclusion

In conclusion, the evaluation shows that the vendor's reported probabilities do not represent a valid probability space. The joint probabilities fail to normalize, the consistency

check reveals a ten percent shortfall, and the conditional statements break the complement rule. These issues mean the dataset cannot satisfy the axioms of probability, making it unreliable for guiding a twenty-million-dollar investment. The key takeaways highlight why this matters: the partition identity ensures that all outcomes in the sample space must balance, the law of total probability requires overall consistency, and conditional probabilities must properly align with their complements. In real-world terms, relying on this dataset could mean rejecting devices that are fully functional or approving defective devices as reliable. Either outcome would create financial loss, increase the risk of warranty claims, and weaken customer confidence in the product. Another implication is that inconsistent testing data could mask deeper issues in quality control, making it difficult for management to identify whether failures come from the devices themselves or from flaws in the testing system. These lessons reinforce that probability is not just a mathematical tool but a framework for making sound decisions. Until the vendor provides corrected values that satisfy the event space and its axioms, my recommendation would reflect as a no-go.

Citations

- ☐ León-García, A. (2009). *Probability, statistics, and random processes for electrical engineering* (Third Edition). Pearson Education.
- ☐ Gill, T. L. (n.d.). EECS-331 Conditional Probability L-4.